DATA MODELS

Learning Objectives

- Explain data modelling and its importance
- Identify components of a data model.

Introduction

- Designers, programmers, and end users see data in different ways.
- Different views of same data lead to designs that do not reflect organization’s operation
- Data modeling reduces complexities of database design

Data Abstraction

- It refers to the suppression of details of data organization and storage, and the highlighting of the essential features for an improved understanding of data.

Data Model

- Model - an abstraction of a real-world object or event
- Data model - a collection of concepts that can be used to describe the structure of a database — provides the necessary means to achieve this abstraction
- Data modelling is iterative and progressive.

The Importance of Data Models

- Communication tool – it facilitate interaction among the designer, the applications programmer, and the end user.
- Well-developed model – even foster improved understanding of the organization.
- Applications are developed to manage data and help transform it into information.

Data Model Basic Building Blocks

- Entity : anything or event about which data are to be collected and stored (Unique &distinct).
- Attribute : a characteristic of an entity.
- Relationship : describes an association among entities.
    - One-to-many (1:M) relationship
    - Many-to-many (M:N or M:M) relationship
    - One-to-one (1:1) relationship
- Constraint : a restriction placed on the data.

Business Rules

- Brief, precise, and unambiguous description of a policy, procedure, or principle
- Enable defining the basic building blocks
- Describe main and distinguishing characteristics of the data

Sources of Business Rules

Company Managers

Policy makers

Department managers

Written documentation

Direct interviews with end users

Reasons for Identifying and Documenting Business Rules

- Help standardize company's view of data
- Communications tool between users and designers
- Allow designer to:
  - Understand the nature, role, scope of data, and business processes
  - Develop appropriate relationship participation rules and constraints
  - Create an accurate data model

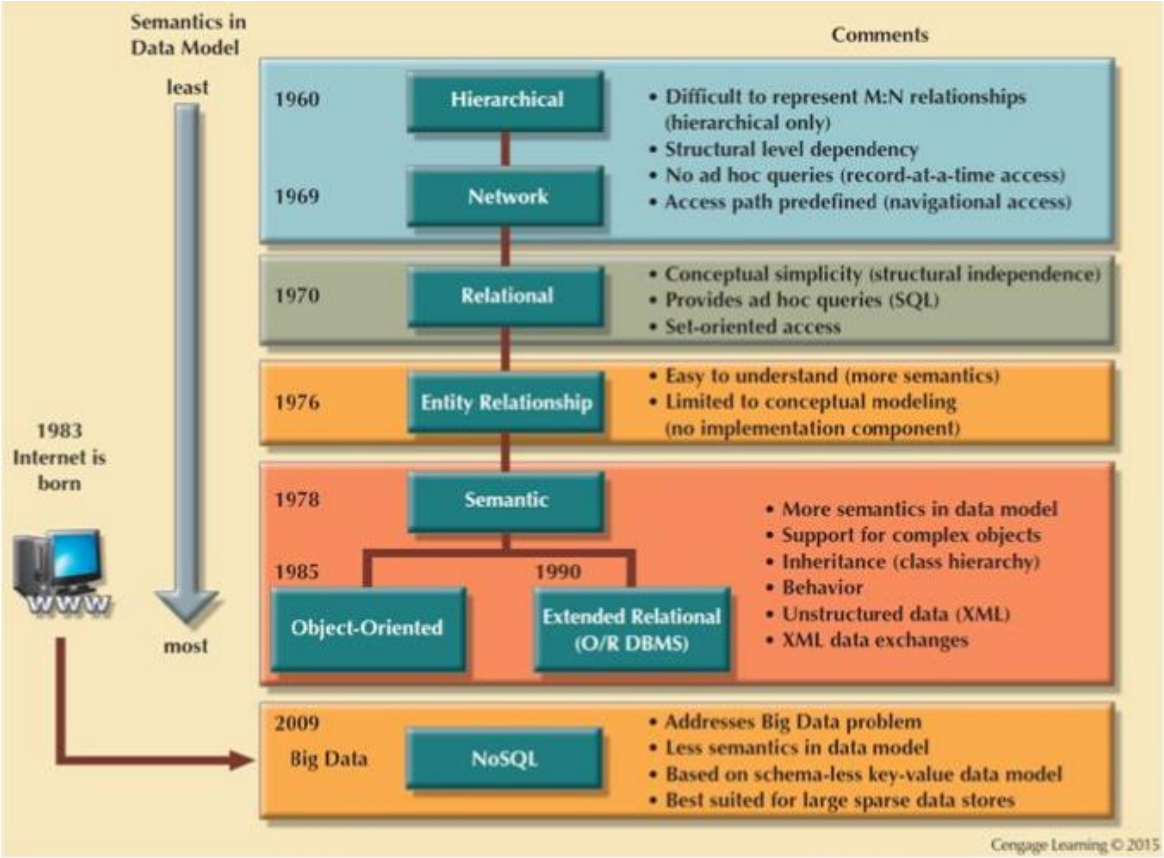
Translating Business Rules into Data Model Components

- Nouns translate into entities
- Verbs translate into relationships among entities
- Relationships are bidirectional
- Questions to identify the relationship type
  - How many instances of B are related to one instance of A?
  - How many instances of A are related to one instance of B?

Naming Conventions Entity names

- Entity names – required to:
  - Be descriptive of the objects in the business environment
  - Use terminology that is familiar to the users
- Attribute name - Required to be descriptive of the data represented by the attribute
- Proper naming:
  - Facilitates communication between parties
  - Promotes self-documentation

The Evolution of Data Models



Hierarchical Model

Advantages	Disadvantages
▪ Promotes data sharing ▪ Parent/child relationship promotes conceptual simplicity and data integrity ▪ Database security is provided and enforced by DBMS ▪ Efficient with 1:M relationships	▪ Requires knowledge of physical data storage characteristics ▪ Navigational system requires knowledge of hierarchical path ▪ Changes in structure require changes in all application programs ▪ Implementation limitations ▪ No data definition ▪ Lack of standards

Network Model

Advantages	Disadvantages
▪ Conceptual simplicity ▪ Handles more relationship types ▪ Data access is flexible ▪ Data owner/member relationship promotes data integrity ▪ Conformance to standards ▪ Includes data definition language (DDL) and data manipulation language (DML)	▪ System complexity limits efficiency ▪ Navigational system yields complex implementation, application development, and management ▪ Structural changes require changes in all application programs

Hierarchical and Network Models

Hierarchical Models	Network Models
▪ Manage large amounts of data for complex manufacturing projects ▪ Represented by an upside-down tree which contains segments ▪ **Segments**: Equivalent of a file system's record type ▪ Depicts a set of one-to-many (1:M) relationships	▪ Represent complex data relationships ▪ Improve database performance and impose a database standard ▪ Depicts both one-to-many (1:M) and many-to-many (M:N) relationships

The Relational Model

- Produced an automatic transmission database that replaced standard transmission databases
- Based on a relation

  Relation or table: Matrix composed of intersecting tuple and attribute

  - Tuple: Rows
  - Attribute: Columns
- Describes a precise set of data manipulation constructs

The Relational Model

Advantages	Disadvantages
▪ Structural independence is promoted using independent tables ▪ Tabular view improves conceptual simplicity ▪ Ad hoc query capability is based on SQL ▪ Isolates the end user from physical-level details ▪ Improves implementation and management simplicity	▪ Requires substantial hardware and system software overhead ▪ Conceptual simplicity gives untrained people the tools to use a good system poorly ▪ May promote information problems

Relational Database Management System (RDBMS)

- Performs basic functions provided by the hierarchical and network DBMS systems
- Makes the relational data model easier to understand and implement
- Hides the complexities of the relational model from the user

The Entity Relationship Model

Advantages	Disadvantages
▪ Visual modeling yields conceptual simplicity ▪ Visual representation makes it an effective communication tool ▪ Is integrated with the dominant relational model	▪ Limited constraint representation ▪ Limited relationship representation ▪ No data manipulation language ▪ Loss of information content occurs when attributes are removed from entities to avoid crowded displays

The Object-Oriented Data Model (OODM) or Semantic Data Model

- Object-oriented database management system(OODBMS)
- Object: Contains data and their relationships with operations that are performed on it

  Basic building block for autonomous structures

  Abstraction of real-world entity
- Attributes - Describe the properties of an object

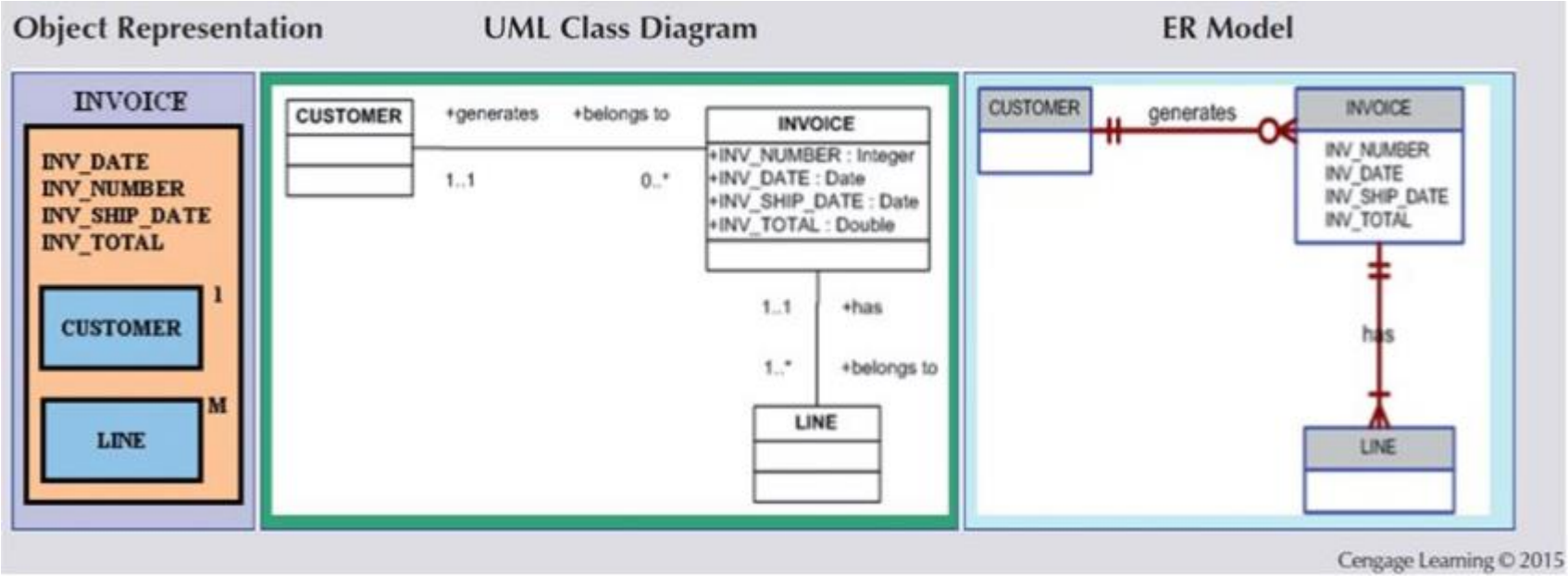
The Object-Oriented Data Model (OODM)

- Class: Collection of similar objects with shared structure and behavior organized in a class hierarchy
  - Class hierarchy: Resembles an upside-down tree in which each class has only one parent
- Inheritance: Object inherits methods and attributes of parent class
- Unified Modeling Language (UML)
  - Describes sets of diagrams and symbols to graphically model a system

The Object-Oriented Data Model (OODM)

Advantages	Disadvantages
▪ Semantic content is added ▪ Visual representation includes semantic content ▪ Inheritance promotes data integrity	▪ Slow development of standards caused vendors to supply their own enhancements ▪ Compromised widely accepted standard ▪ Complex navigational system ▪ Learning curve is steep ▪ High system overhead slows transactions

A Comparison of OO, UML, and ER Models



Object/Relational and XML

- Extended relational data model (ERDM)
  - Supports OO features and complex data representation
  - Object/Relational Database Management System (O/R DBMS)
  - Based on ERDM, focuses on better data management
- Extensible Markup Language (XML)
  - Manages unstructured data for efficient and effective exchange of all data types

Big Data

- Aims to:
  - Find new and better ways to manage large amounts of web and sensor-generated data
  - Provide high performance and scalability at a reasonable cost
- Characteristics
  - Volume
  - Velocity
  - Variety

Big Data Challenges

- Volume does not allow the usage of conventional structures
- Expensive
- OLAP tools proved inconsistent dealing with unstructured data

Big Data New Technologies

Hadoop

Hadoop Distributed File System (HDFS)

MapReduce

NoSQL

NoSQL Databases

- Not based on the relational model
- Support distributed database architectures
- Provide high scalability, high availability, and fault tolerance
- Support large amounts of sparse data
- Geared toward performance rather than transaction consistency
- Store data in key-value stores

NoSQL

Advantages	Disadvantages
▪ High scalability, availability, and fault tolerance are provided ▪ Uses low-cost commodity hardware ▪ Supports Big Data ▪ 4. Key-value model improves storage efficiency	▪ Complex programming is required ▪ There is no relationship support ▪ There is no transaction integrity support ▪ In terms of data consistency, it provides an eventually consistent model

Data Model Basic Terminology Comparison

REAL WORLD	EXAMPLE	FILE PROCESSING	HIERARCHICAL MODEL	NETWORK MODEL	RELATIONAL MODEL	ER MODEL	OO MODEL
A group of vendors	Vendor file cabinet	File	Segment type	Record type	Table	Entity set	Class
A single vendor	Global supplies	Record	Segment occurrence	Current record	Row (tuple)	Entity occurrence	Object instance
The contact name	Johnny Ventura	Field	Segment field	Record field	Table attribute	Entity attribute	Object attribute
The vendor identifier	G12987	Index	Sequence field	Record key	Key	Entity identifier	Object identifier

Note: For additional information about the terms used in this table, consult the corresponding chapters and online appendixes that accompany this book. For example, if you want to know more about the OO model, refer to Appendix G, Object-Oriented Databases.

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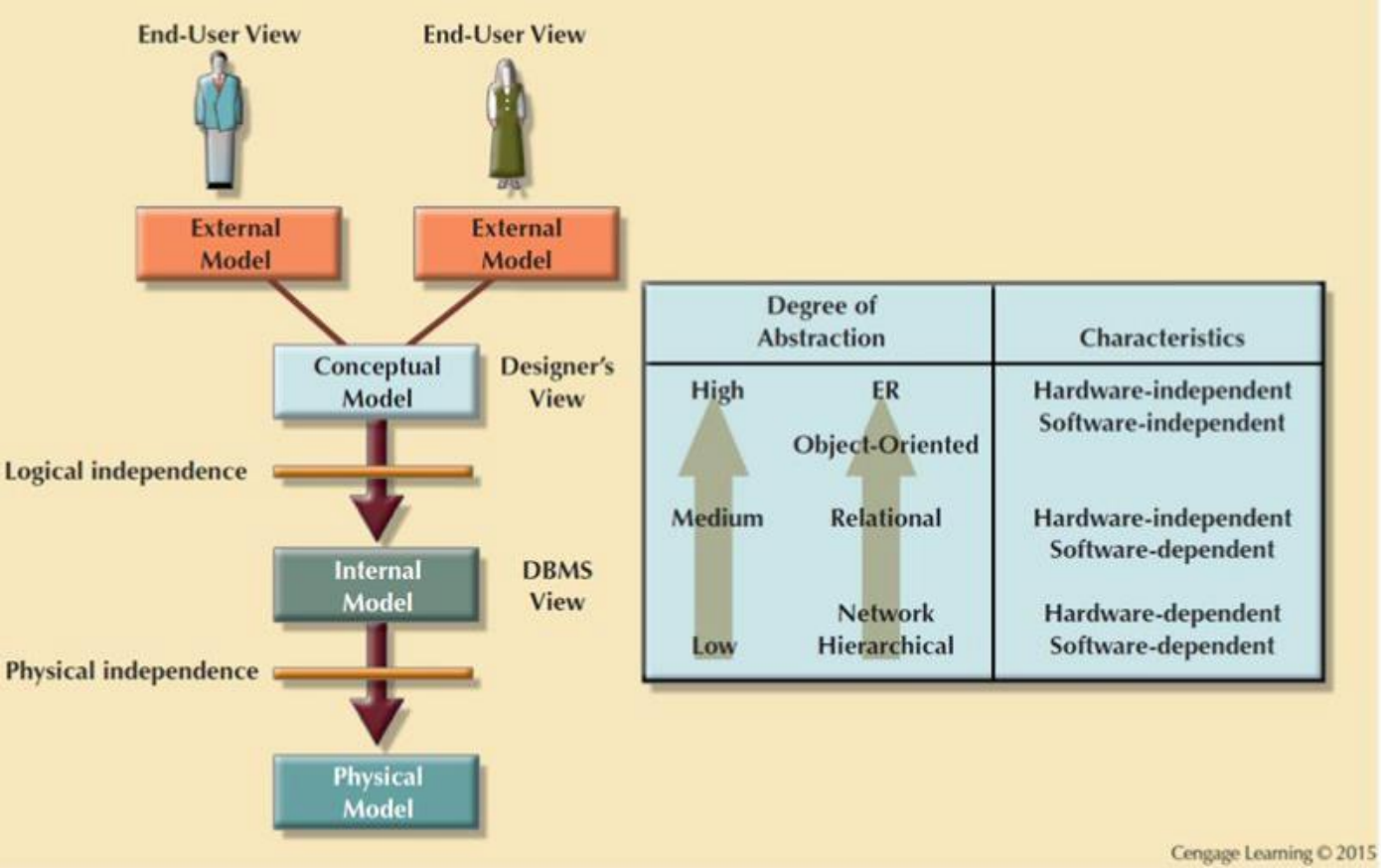
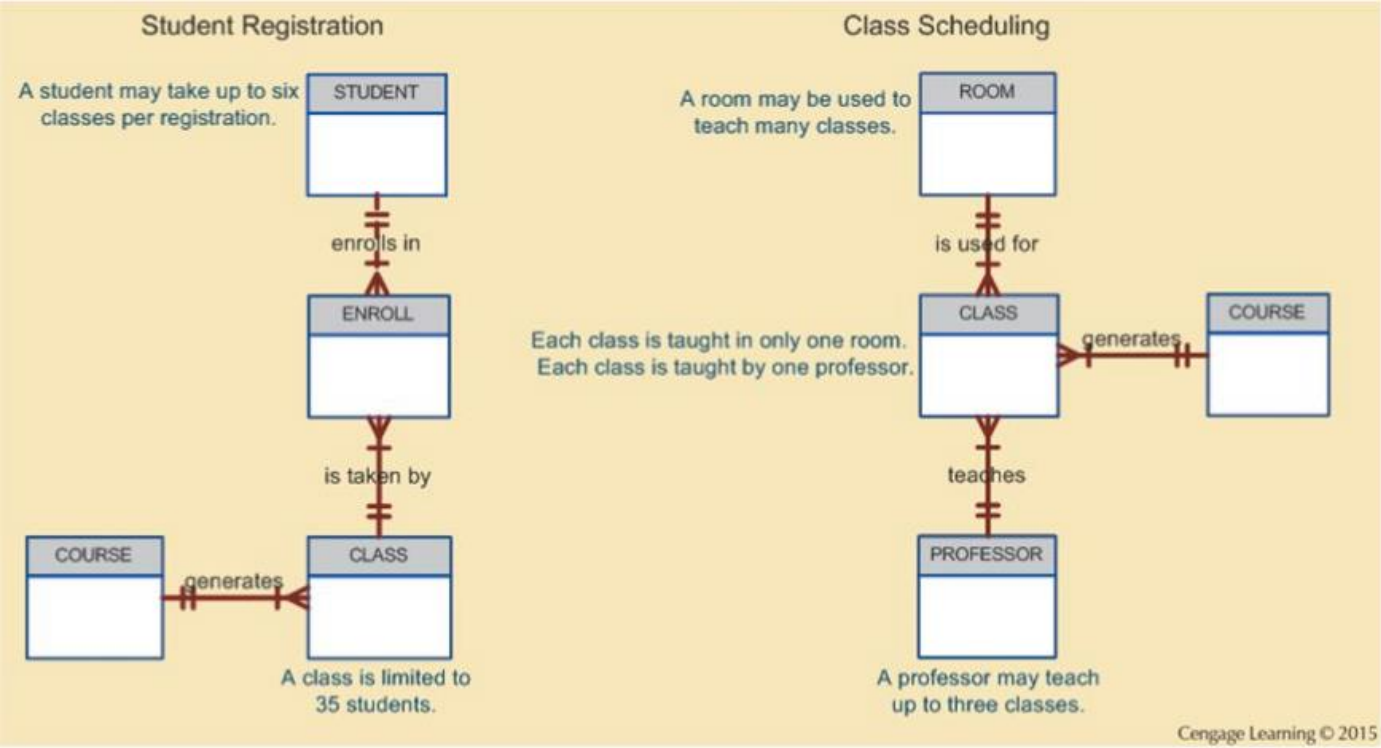
Data Abstraction Levels



The External Model

- End users’ view of the data environment
- ER diagrams are used to represent the external views
- External schema: Specific representation of an external view

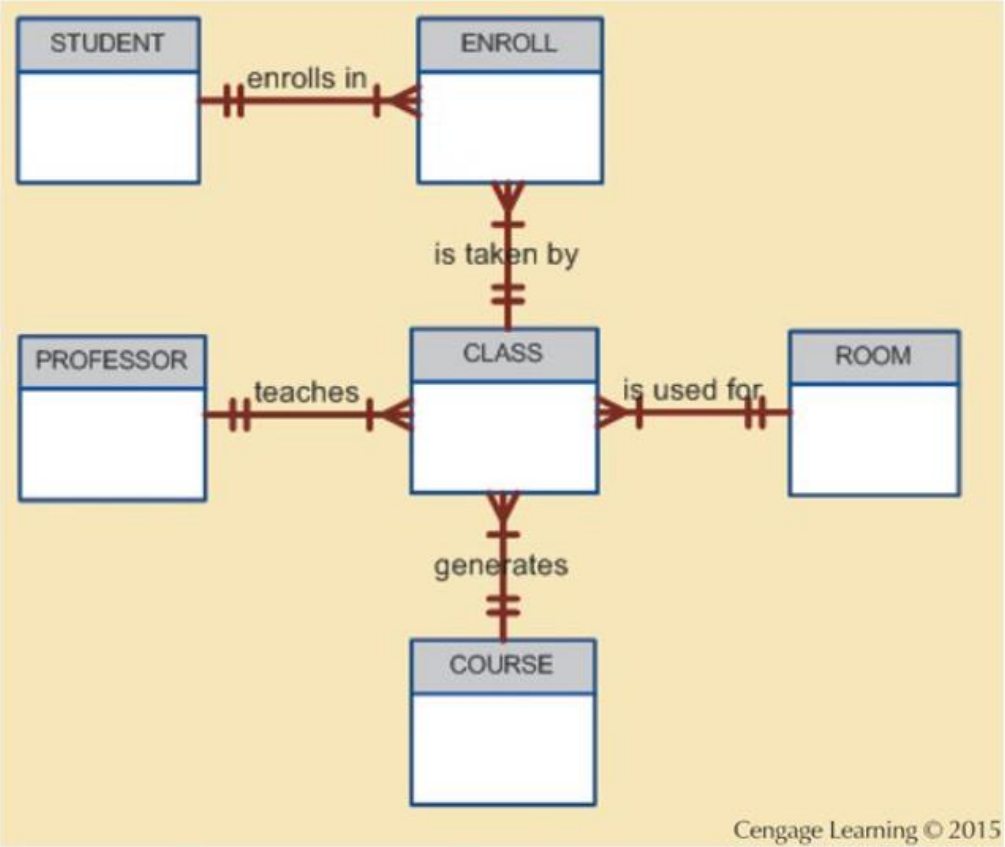
External Models for Tiny College



The Conceptual Model

- Represents a global view of the entire database by the entire organization
- Conceptual schema: Basis for the identification and high-level description of the main data objects
- Has a macro-level view of data environment
- Is software and hardware independent
- Logical design: Task of creating a conceptual data model

The Conceptual Model for Tiny College



The Internal Model

- Representing database as seen by the DBMS mapping conceptual model to the DBMS
- Internal schema: Specific representation of an internal model

  Uses the database constructs supported by the chosen database
- Is software dependent and hardware independent
- Logical independence: Changing internal model without affecting the conceptual model

The Internal Model

- Representing database as seen by the DBMS mapping conceptual model to the DBMS
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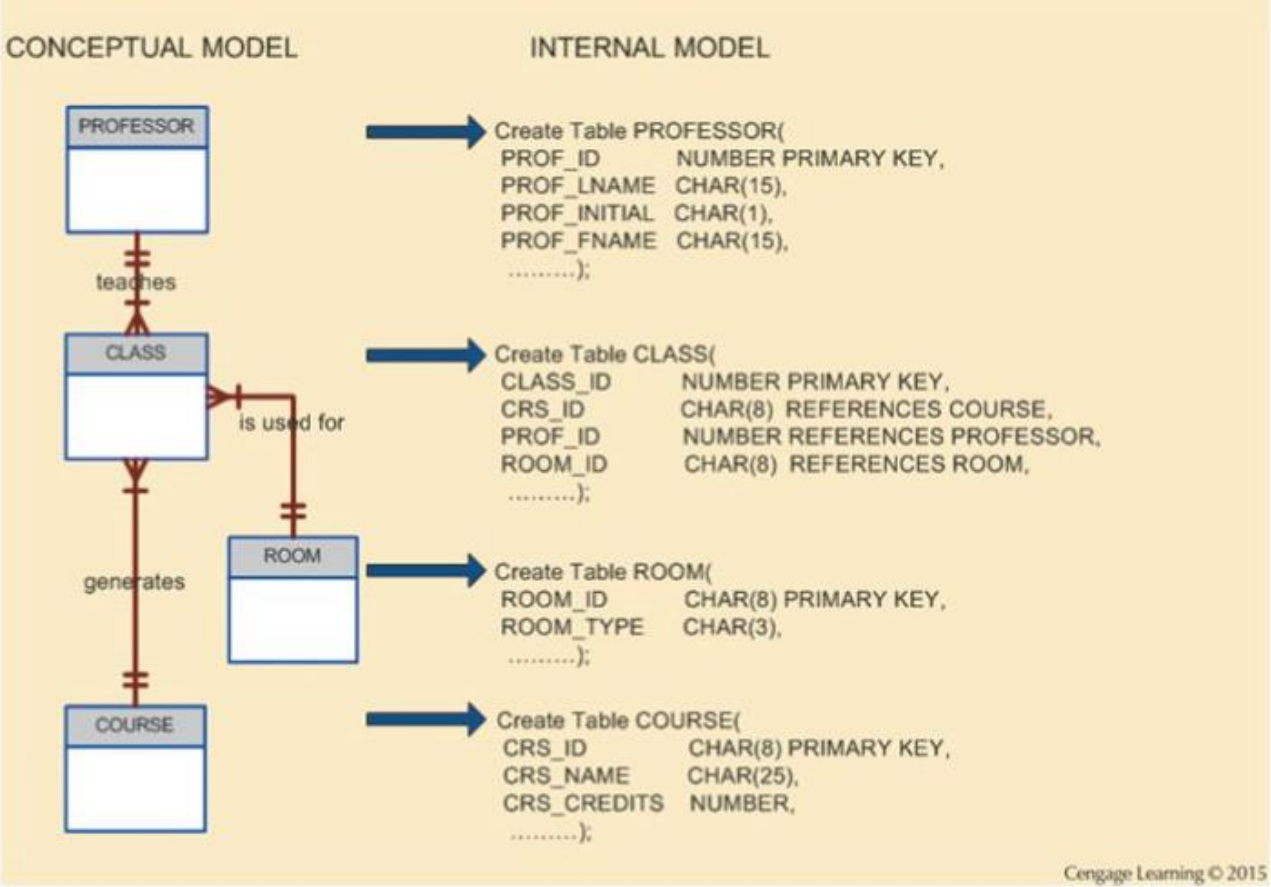
The Internal Model for Tiny College



The Physical Model

- Operates at lowest level of abstraction
- Describes the way data are saved on storage media such as disks or tapes
- Requires the definition of physical storage and data access methods
- Relational model aimed at logical level
- Does not require physical-level details
- Physical independence: Changes in physical model do not affect internal model

Level of Data Abstraction

MODEL	DEGREE OF ABSTRACTION	FOCUS	INDEPENDENT OF
External	High	End-user views	Hardware and software
Conceptual	↕	Global view of data (database model independent)	Hardware and software
Internal		Specific database model	Hardware
Physical	Low	Storage and access methods	Neither hardware nor software

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ER MODEL

BASIC CONCEPTS

Entity

- An entity can be a real-world object, either animate or inanimate, that can be easily identifiable.
- For example (school database):
  - students
  - teachers
  - classes
  - courses offered

Attributes

- Entities are represented by means of their properties, called *attributes.*
- All attributes have values.
- For example a student entity may have

  - name
  - class
  - age

Type of Attributes

- **Simple attributes** – atomic value, which cannot be divided further. E.g. a student's phone number is an atomic value of 10 digits.
- **Composite attributes** – made of more than one simple attribute. E.g. a student's complete name may have first_name and last_name.
- **Derived attributes** – do not exist in the physical database, but their values are derived from other attributes present in the database. E.g. average_salary

Type of Attributes

- **Single-value attributes** – contain single value. E.g. Social_Security_Number.
- **Multi-value attributes** – contain more than one values. E.g. a person can have more than one phone number, email_address, etc.

Relationship

- The association among entities is called a relationship.
- Example:

  an employee **works_at** a department

  a student **enrolls** in a course

Degree of Relationship

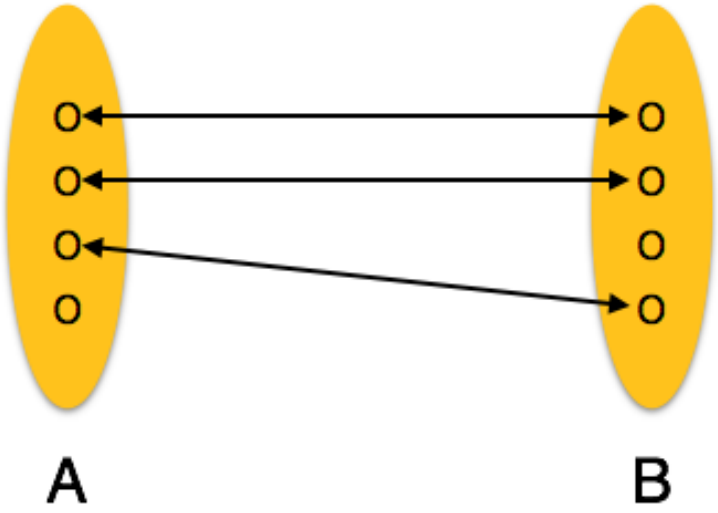
- Binary = degree 2
- Ternary = degree 3
- n-ary = degree

Mapping Cardinalities

- **Cardinality** defines the number of entities in one entity set, which can be associated with the number of entities of other set via relationship set.

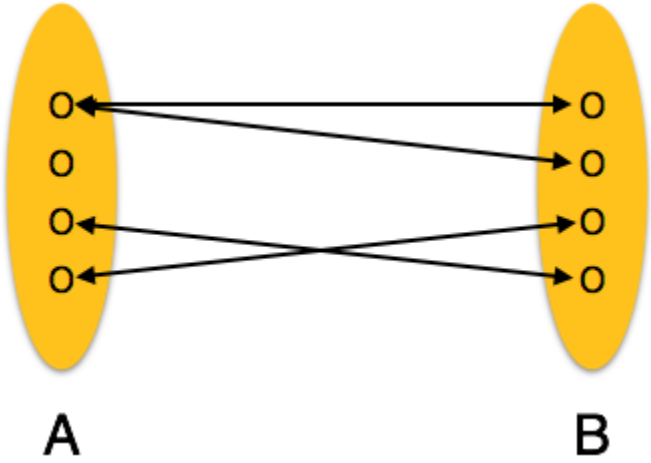
Mapping Cardinalities : ONE-TO-ONE

One entity from entity set A can be associated with at most one entity of entity set B and vice versa.



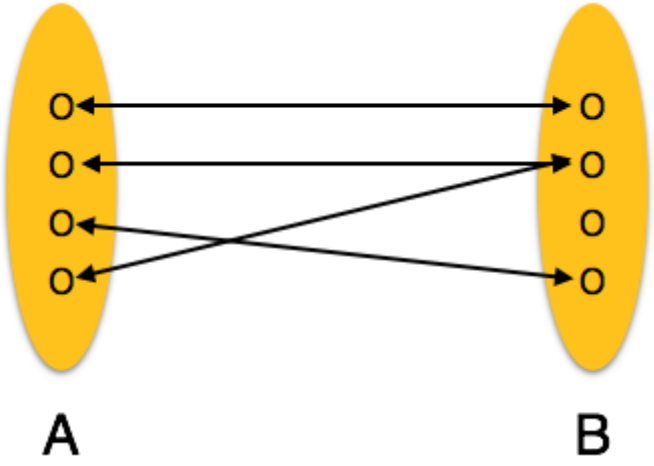
Mapping Cardinalities : ONE-TO-MANY

One entity from entity set A can be associated with more than one entities of entity set B however an entity from entity set B, can be associated with at most one entity.



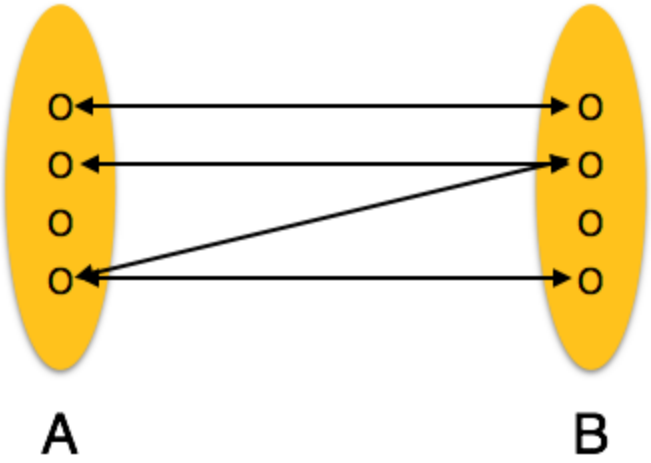
Mapping Cardinalities : MANY-TO-ONE

More than one entities from entity set A can be associated with at most one entity of entity set B, however an entity from entity set B can be associated with more than one entity from entity set A.

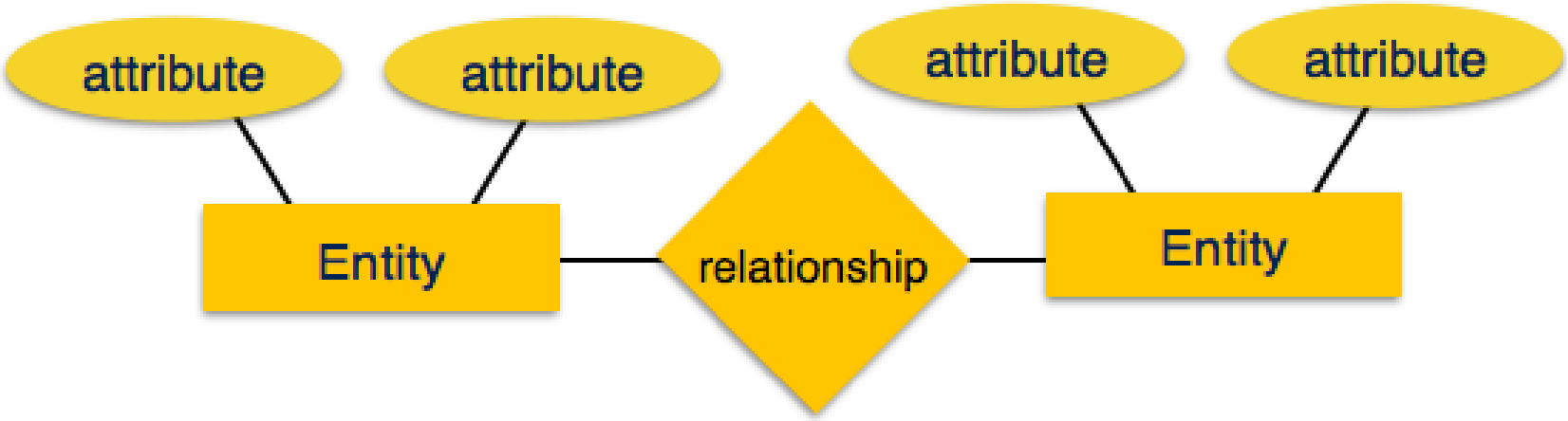


Mapping Cardinalities : MANY-TO-MANY

One entity from A can be associated with more than one entity from B and vice versa.

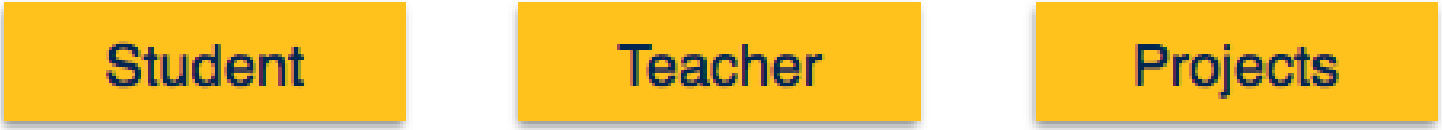


ER DIAGRAM


attribute
attribute
attribute
attribute
Entity
relationship
Entity

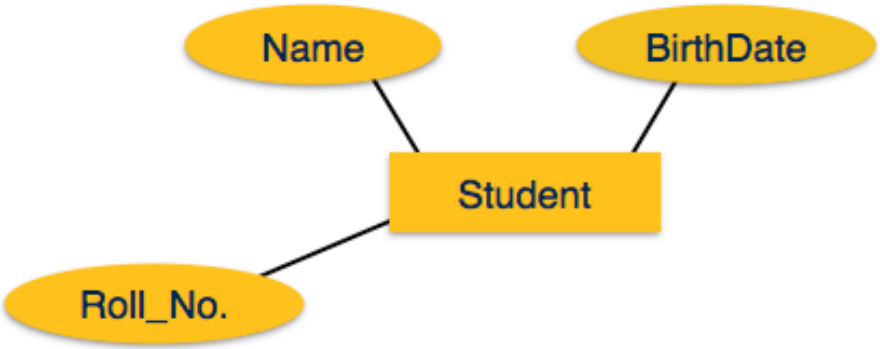
Entity

Entities are represented by means of rectangles.



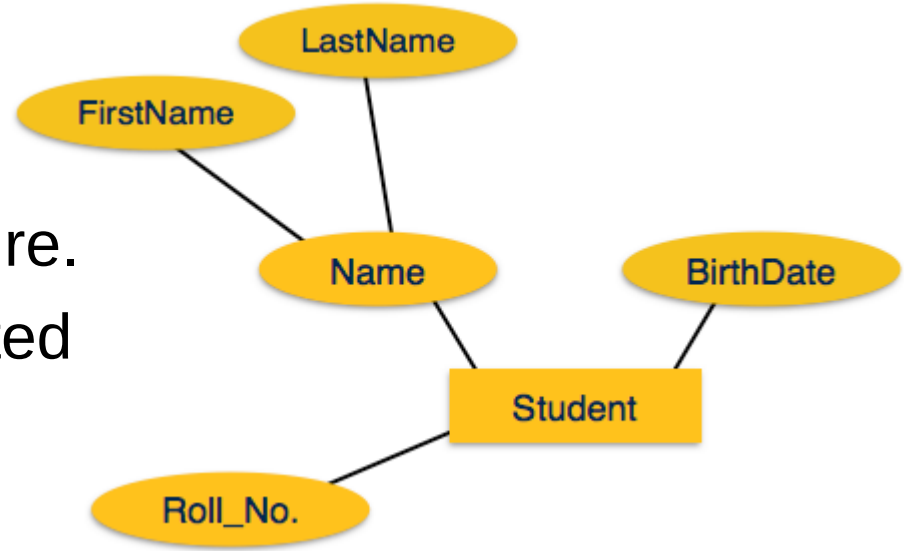
Attributes

- Attributes are the properties of entities.
- Attributes are represented by means of ellipses.
- Every ellipse represents one attribute and is directly connected to its entity (rectangle).



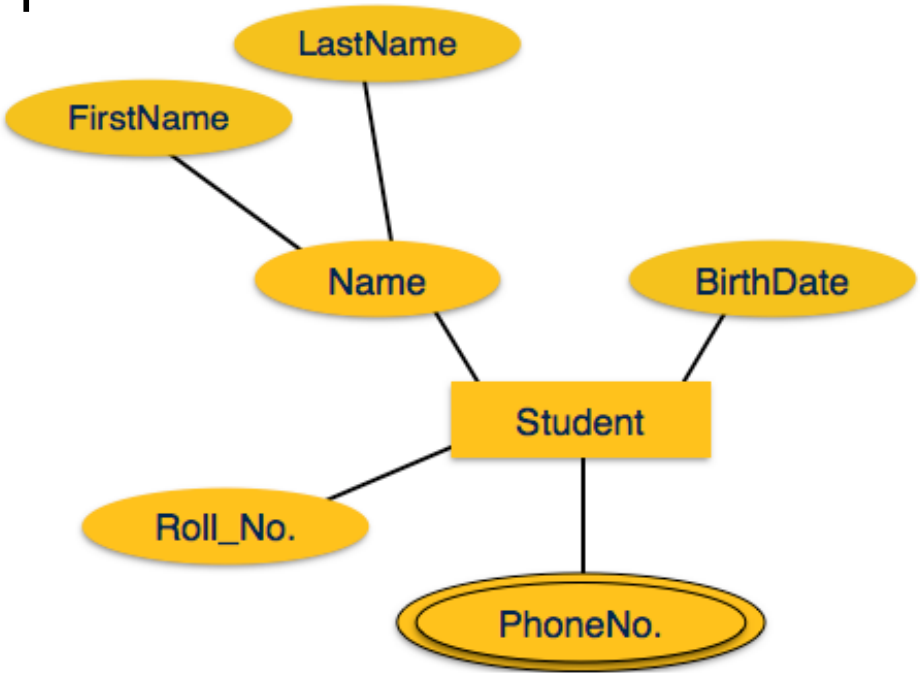
Composite Attributes

- If the attributes are composite, they are further divided in a tree like structure.
- Every node is then connected to its attribute.
- Composite attributes are represented by ellipses that are connected with an ellipse.



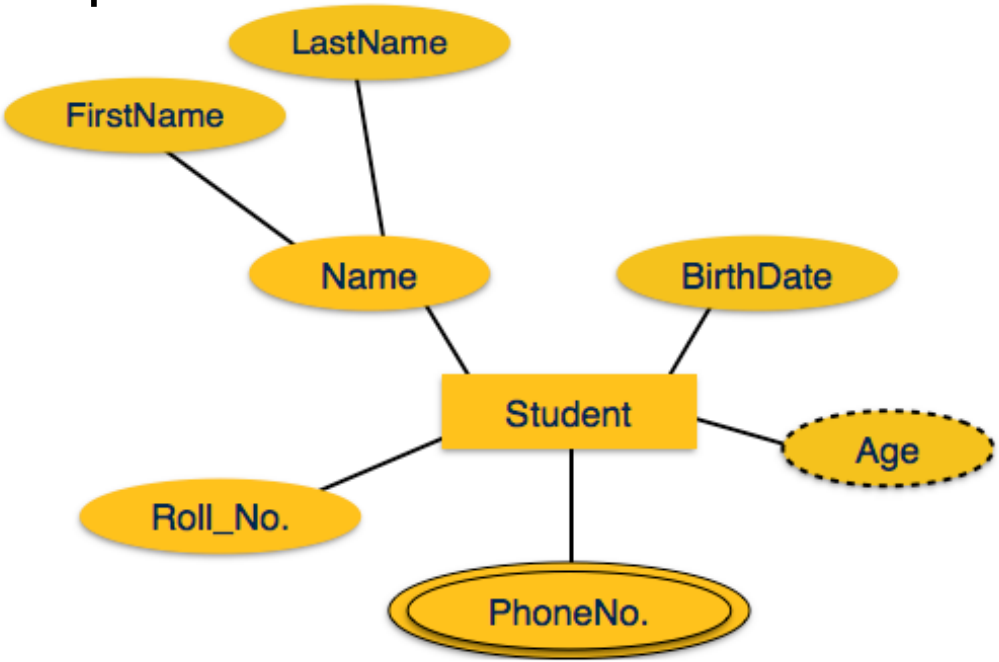
Multivalued Attributes

- Depicted by double ellipse.



Derived Attributes

- Depicted by dashed ellipse.



Relationships

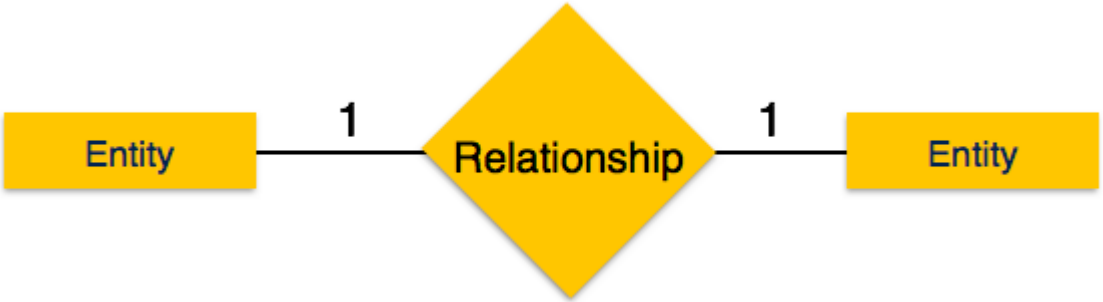
- Entities are represented by means of rectangles.
- Name of the relationship is written inside the diamond-box.
- All the entities (rectangles) participating in a relationship, are connected to it by a line.

Binary Relationship and Cardinality

- Binary relationship - two entities are participating
- Cardinality - the number of instance of an entity from a relation that can be associated with the relation

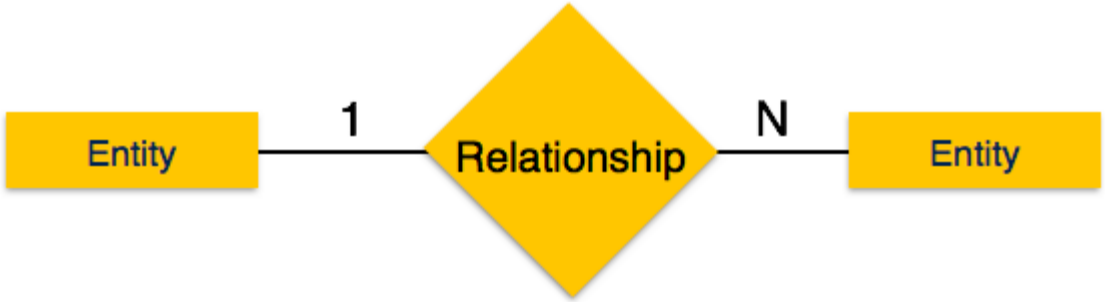
ONE-TO-ONE

- When only one instance of an entity is associated with the relationship, it is marked as '1:1'.



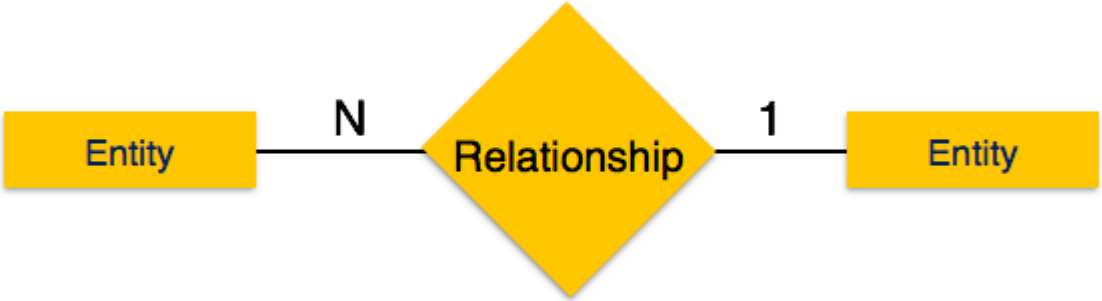
ONE-TO-MANY

- When more than one instance of an entity is associated with a relationship, it is marked as '1:N'.

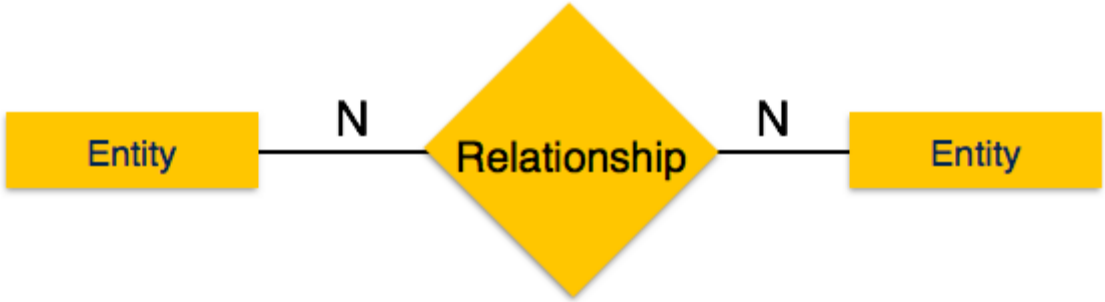


MANY-TO-ONE

- When more than one instance of entity is associated with the relationship, it is marked as 'N:1'.

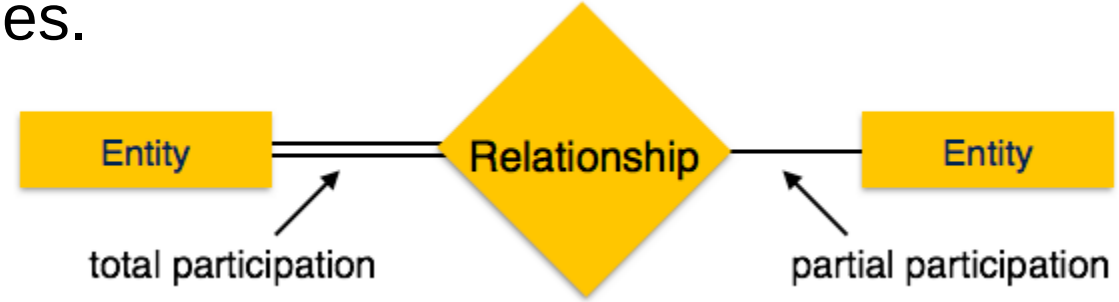


MANY-TO-MANY

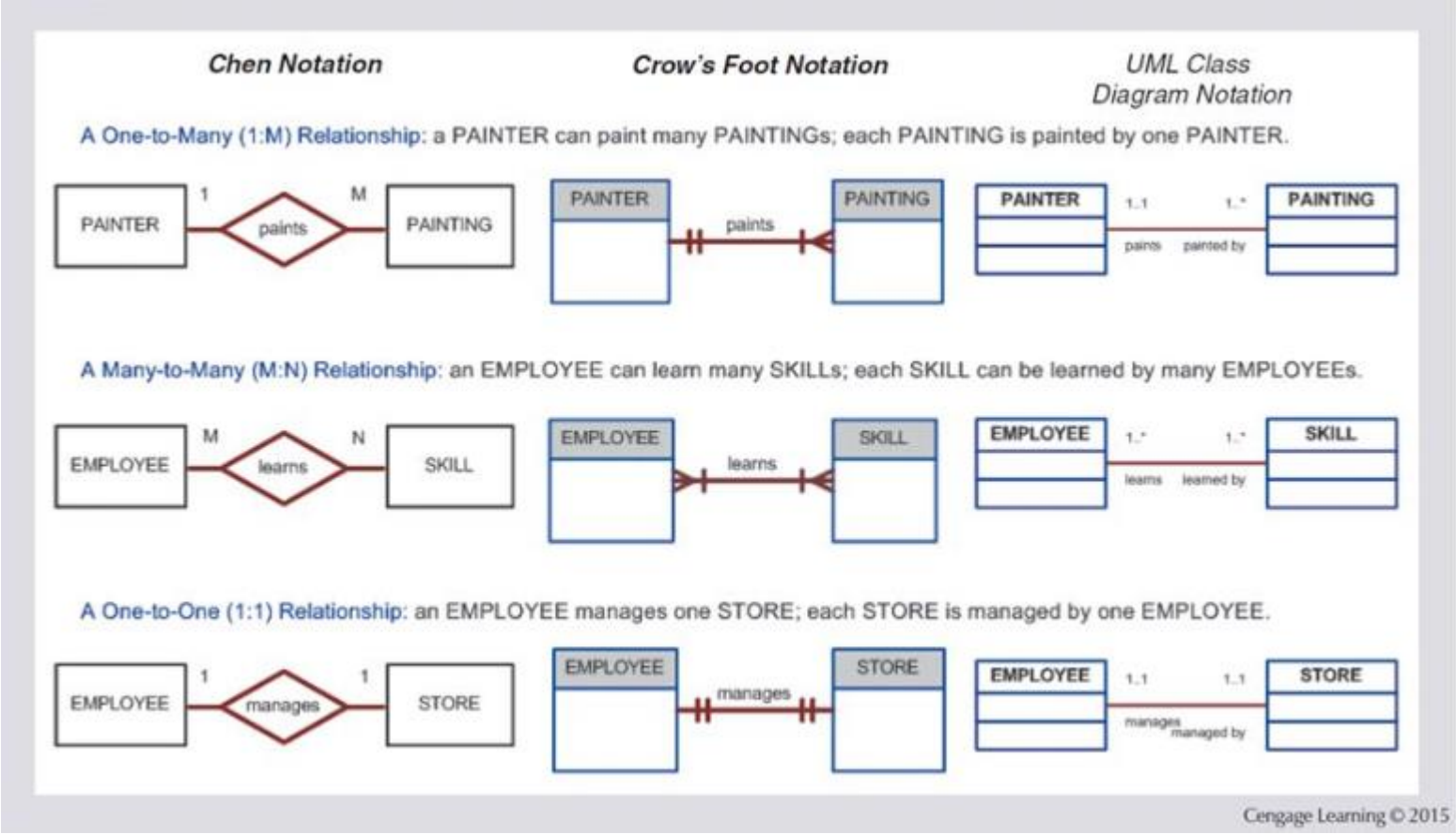


Participation Constraints

- Total Participation − Each entity is involved in the relationship. Total participation is represented by double lines.
- Partial participation − Not all entities are involved in the relationship. Partial participation is represented by single lines.



The ER Model Notations



Thanks!

Any questions?